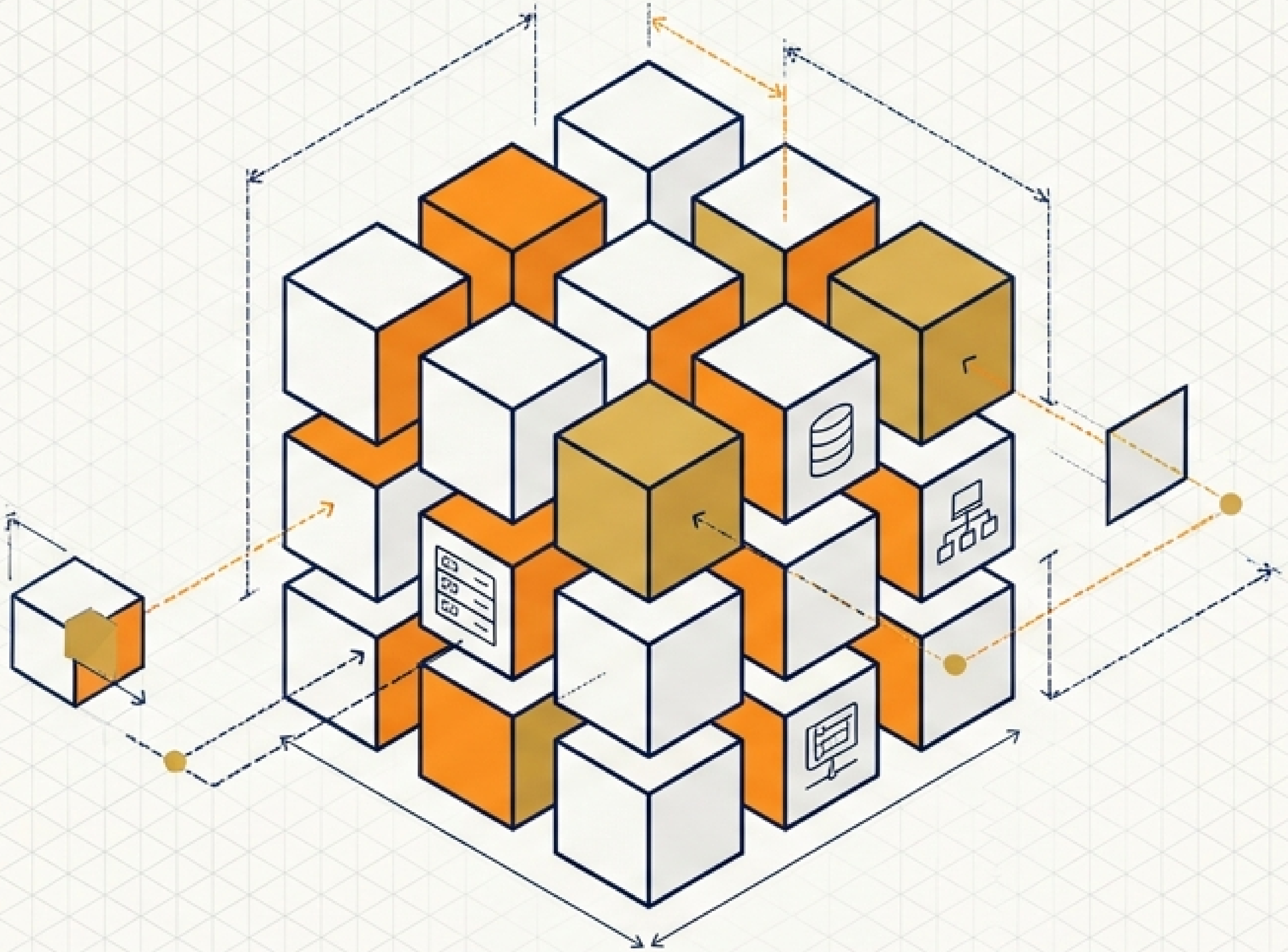
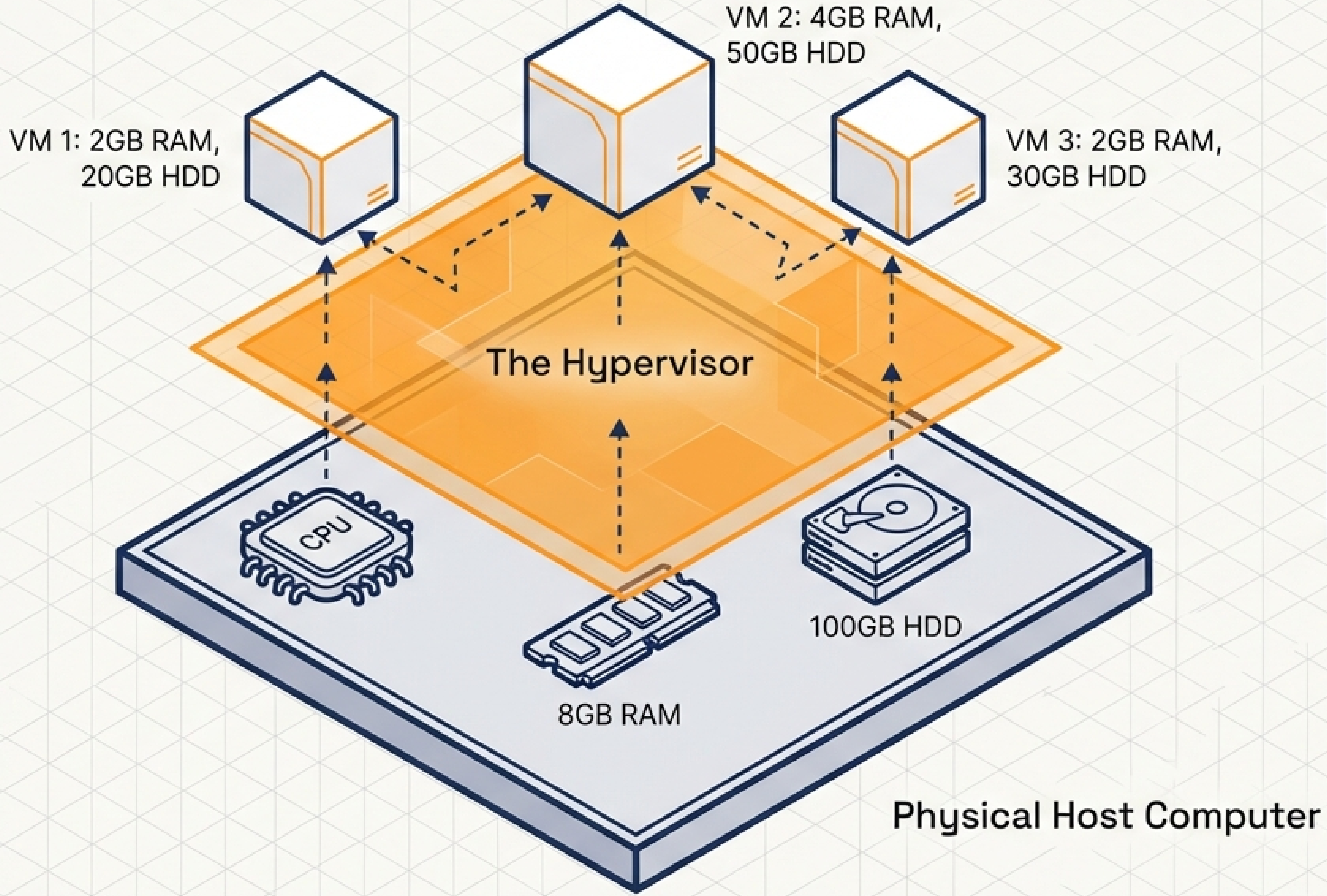


The Architect's Building Blocks

Demystifying Cloud Computing & AWS Infrastructure

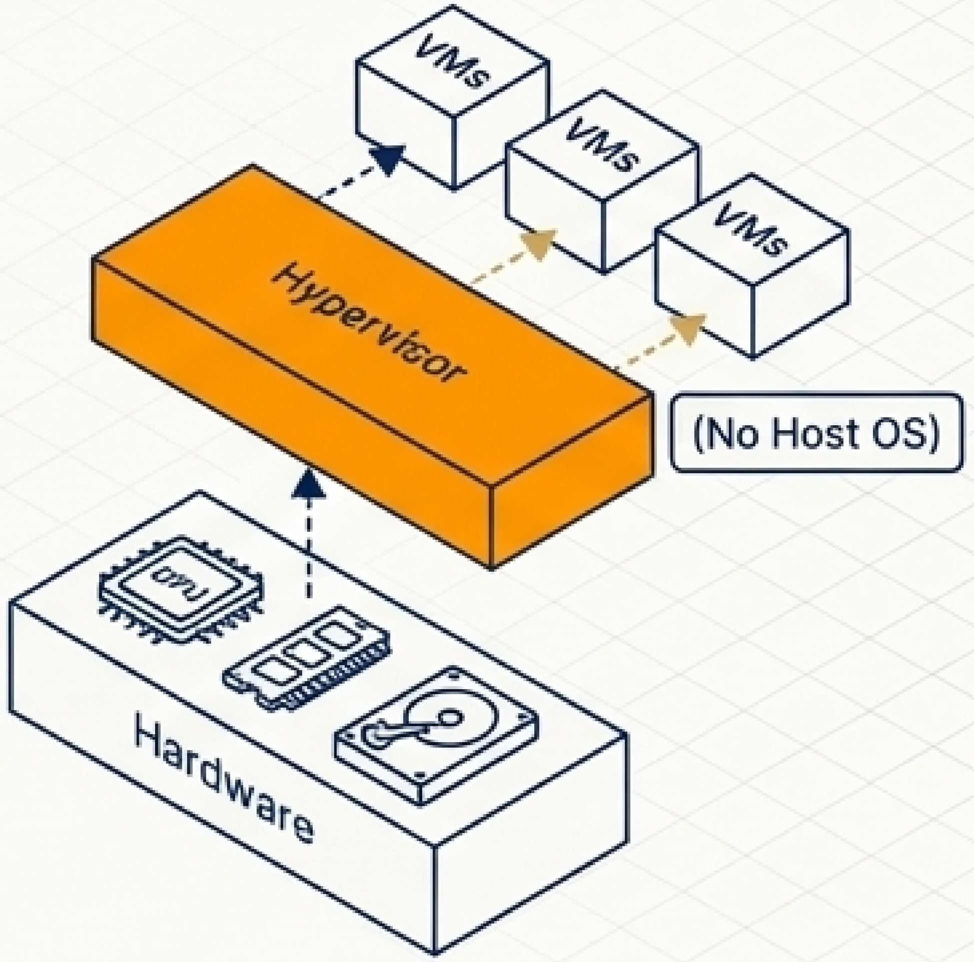
The Virtualization Engine



A Hypervisor is a software layer that slices physical hardware into isolated, independent mini-computers (VMs), allowing multiple operating systems to run simultaneously on a single machine.

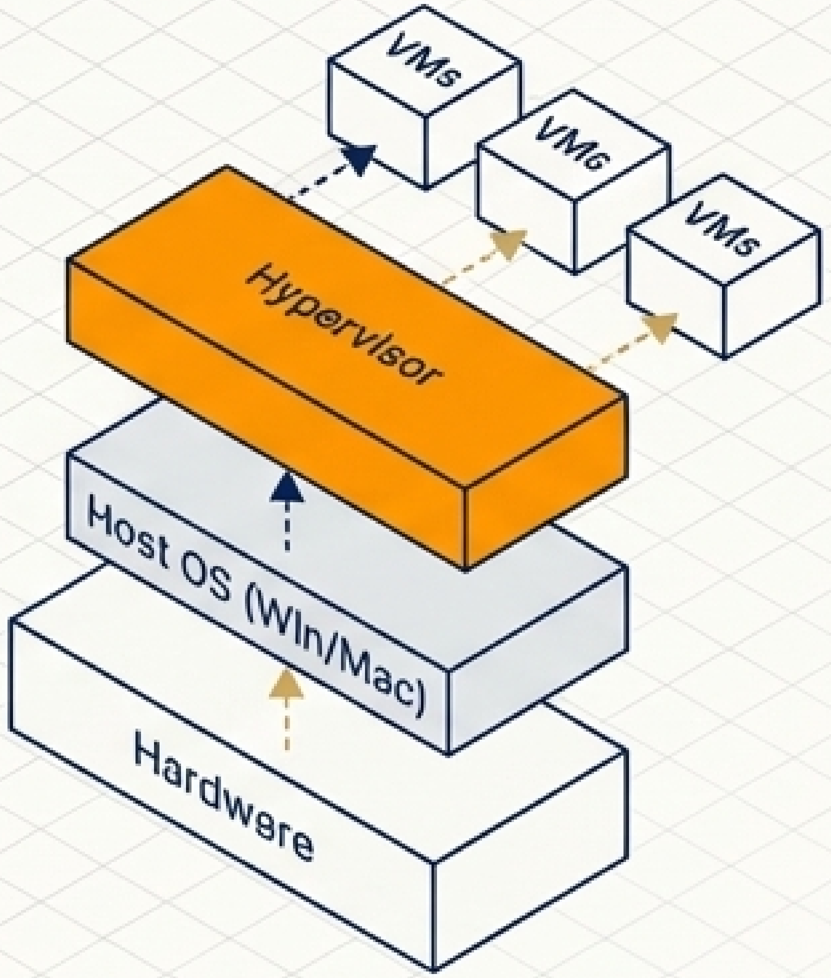
The Hypervisor Matrix

Type 1 (Bare Metal)



- **Setup:** Installed directly on physical hardware like a mini-OS.
- **Use Case:** Enterprise Data Centers & Cloud Providers (AWS, VMware). High performance, scalable.

Type 2 (Hosted)

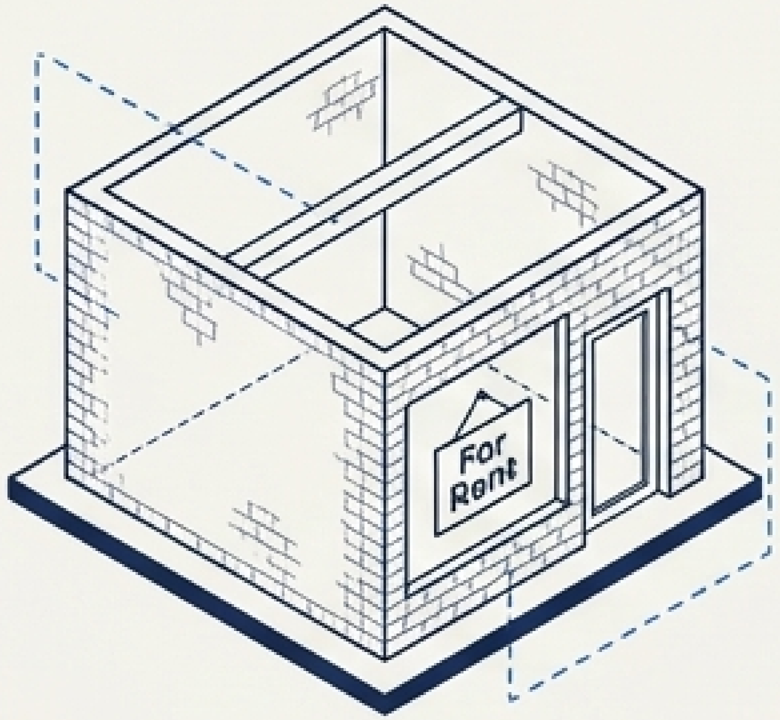


- **Setup:** Installed as an application on an existing operating system.
- **Use Case:** Personal testing and local development (Oracle VirtualBox). Easy to set up, but shares overhead with Host OS.

The Gradient of Cloud Responsibility

User Responsibility ⟵⟶ Provider Responsibility

Infrastructure as a Service (IaaS)

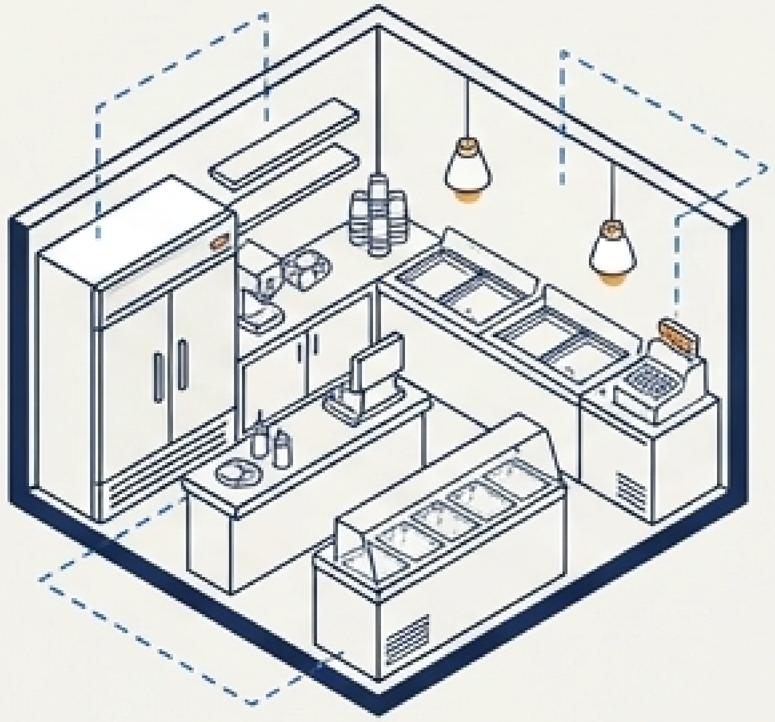


The Empty Shop

Renting the bare physical space. You build the furniture, paint the walls, and bring the ice cream machine.

Cloud Equivalent: You get a blank virtual computer (EC2). You manage the OS, software, and security.

Platform as a Service (PaaS)

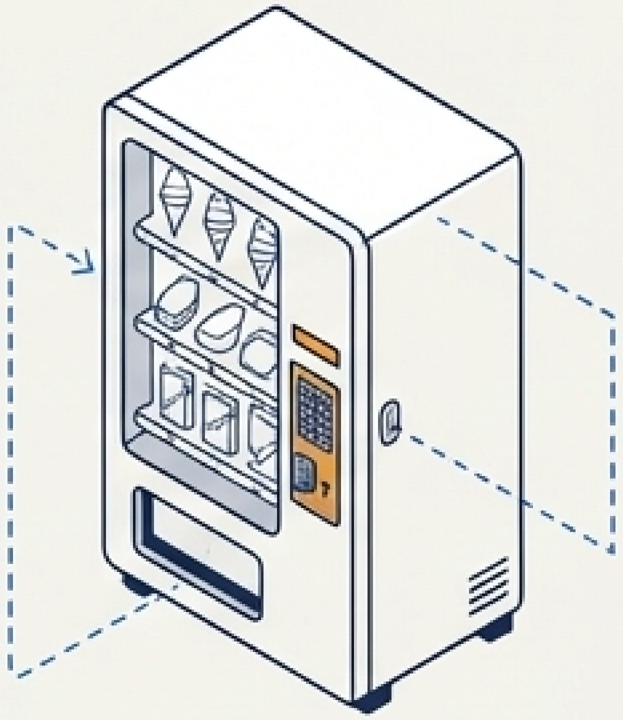


The Furnished Parlor

The shop is ready. You just bring your secret ice cream recipe and start selling.

Cloud Equivalent: The runtime environment is managed. You focus entirely on deploying your application code.

Software as a Service (SaaS)



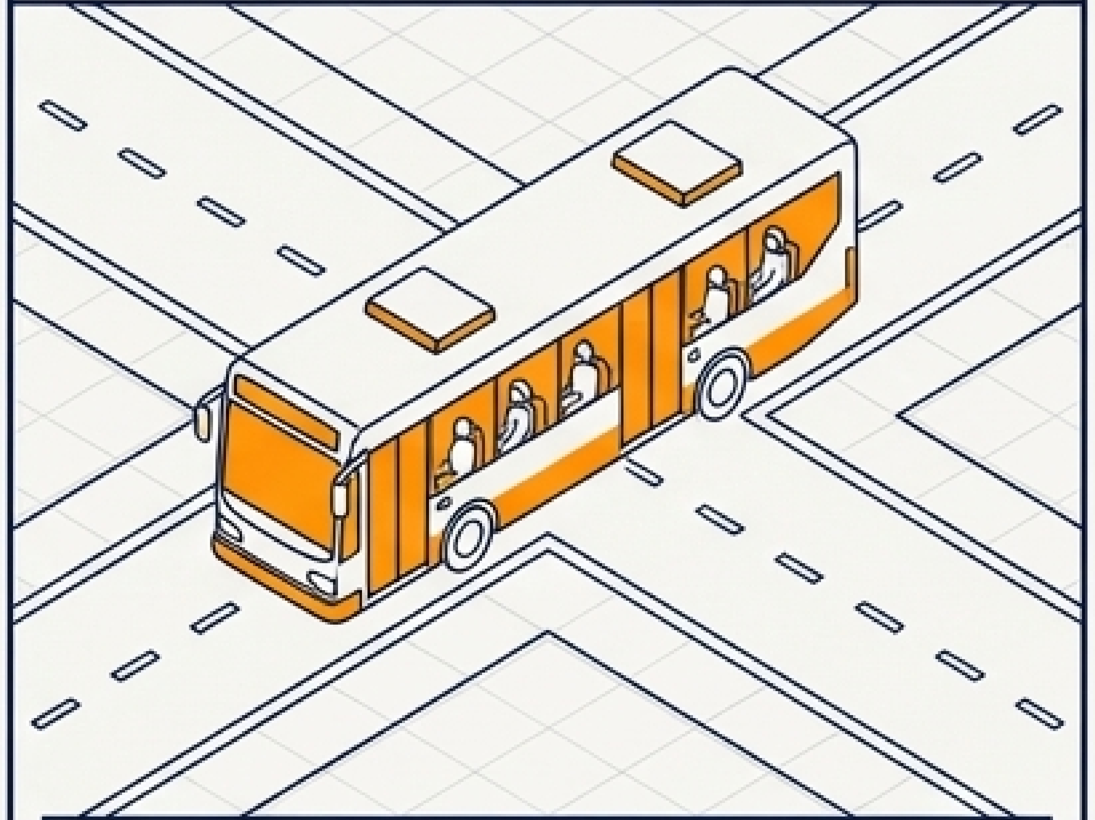
The Vending Machine

Users just insert a coin and get the product. No setup, no maintenance, no management.

Cloud Equivalent: Fully managed internet software (Google Drive, Zoom).

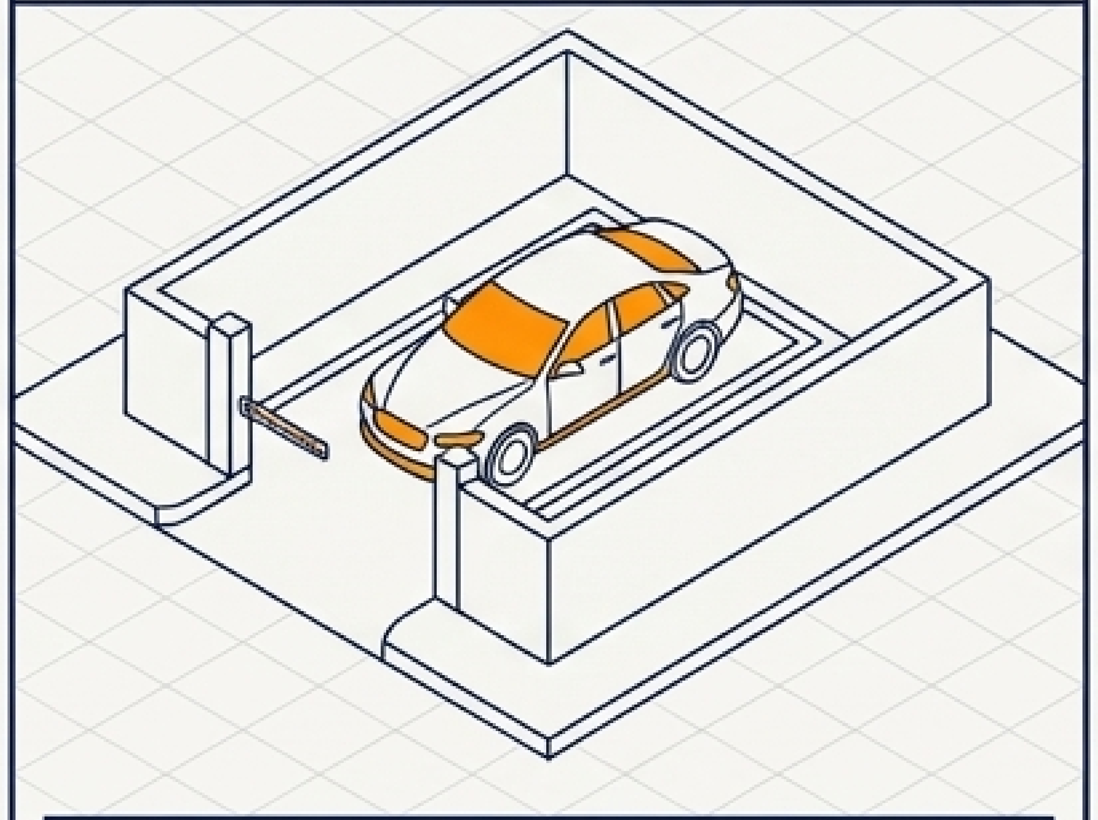
Choosing a Deployment Strategy

Public Cloud
(The Bus)



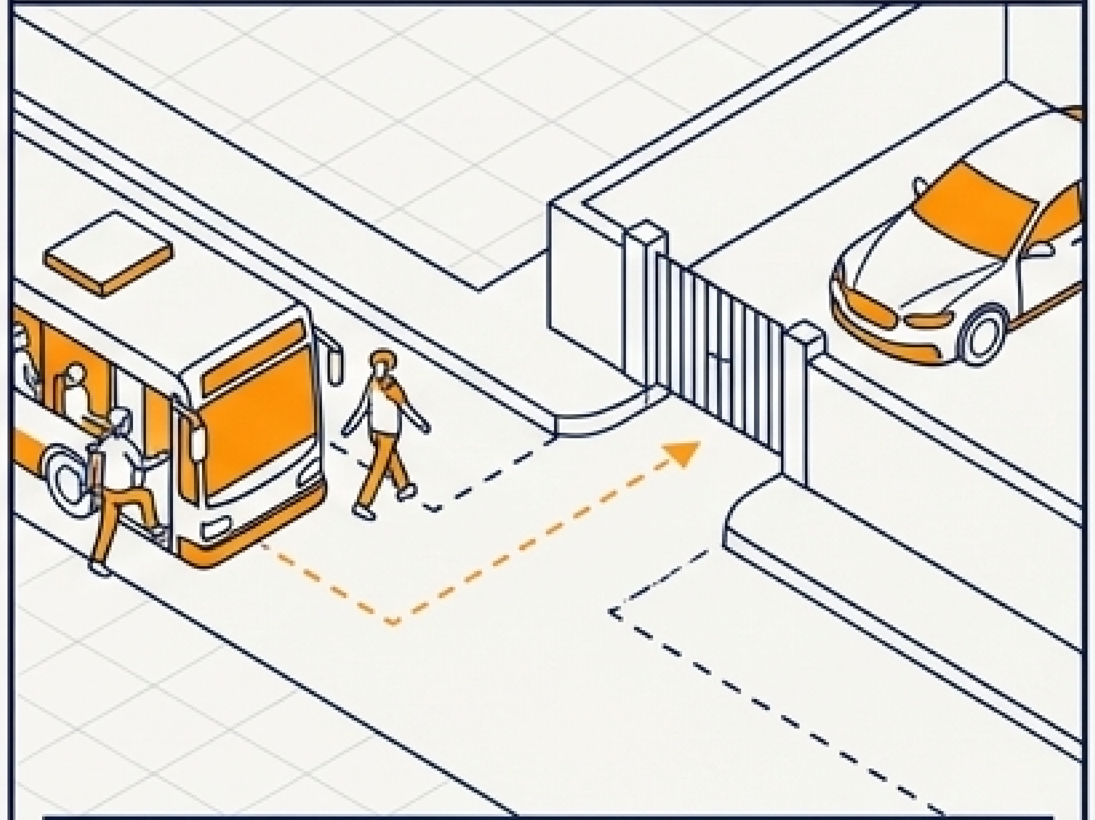
Shared environment where multiple users rent space on massive provider-owned servers (AWS, Azure). Highly cost-effective, zero hardware maintenance, but shared underlying infrastructure.

Private Cloud
(The Rental Car)



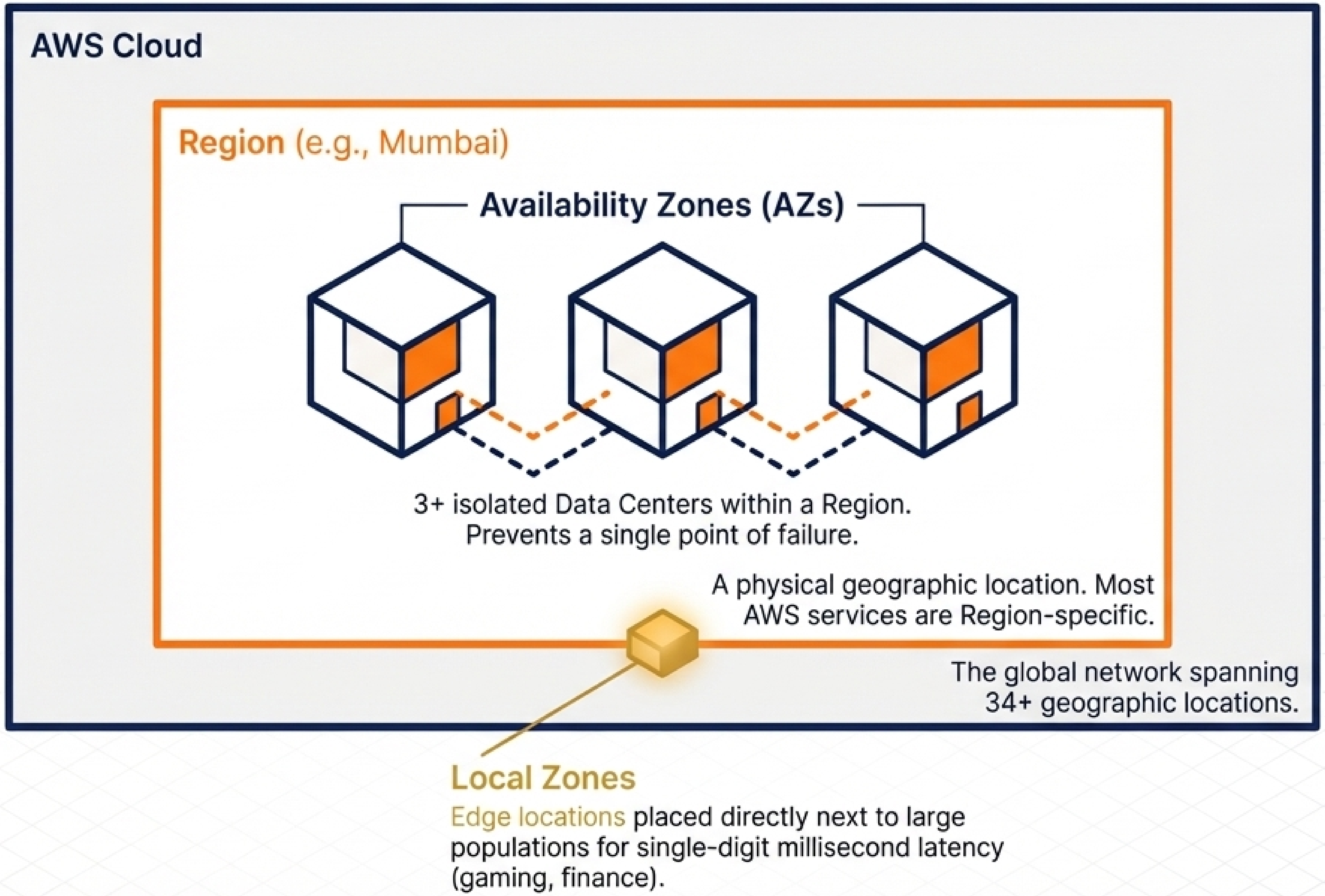
Dedicated cloud environment for a single organization. Offers total control and maximum privacy, but at a significantly higher cost and management overhead.

Hybrid Cloud
(The Mixed Commute)

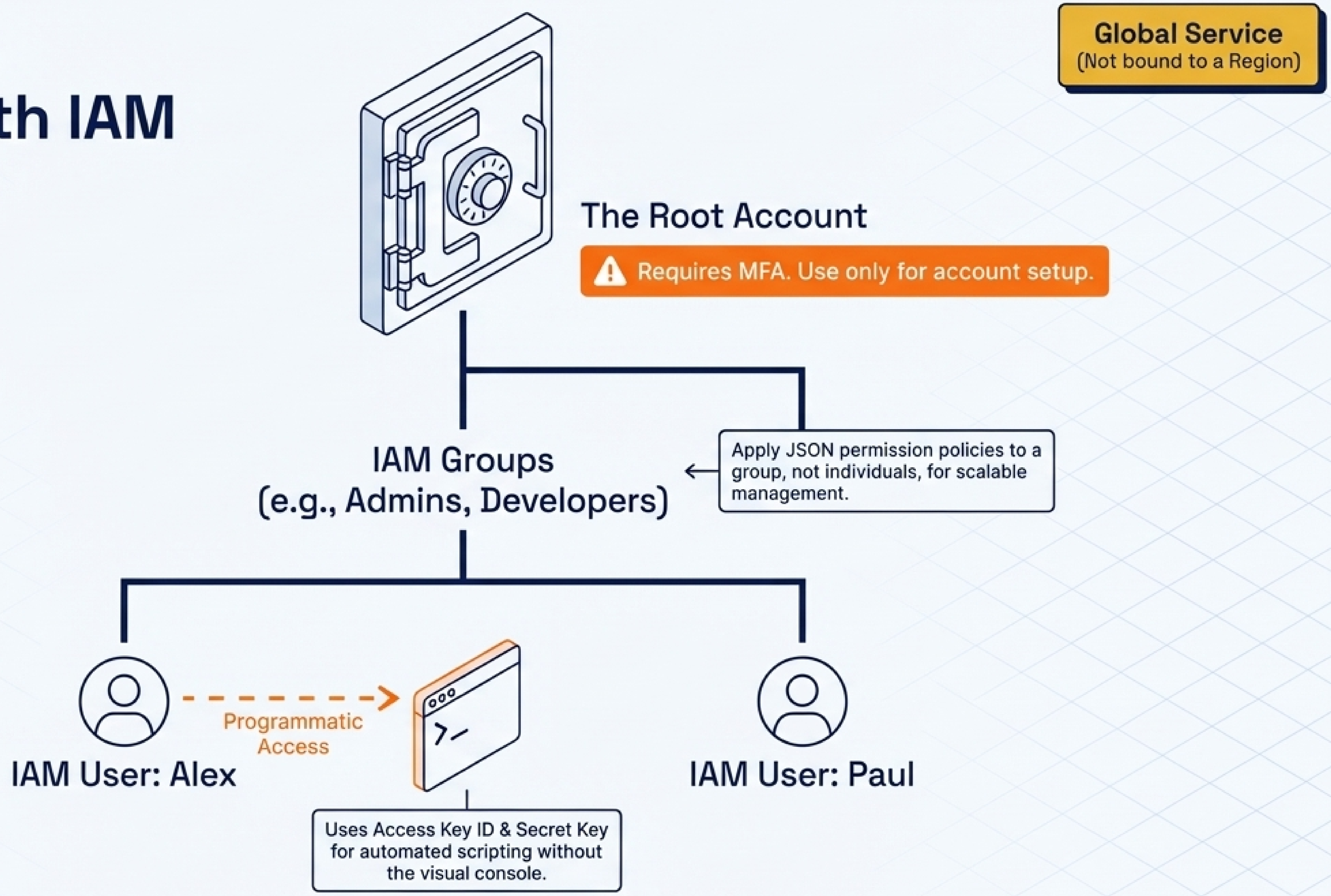


A mix of both. Keep sensitive core data on a Private Cloud while utilizing Public Cloud infrastructure for heavy, public-facing web traffic.

The Global Infrastructure Matryoshka



Securing the Perimeter with IAM



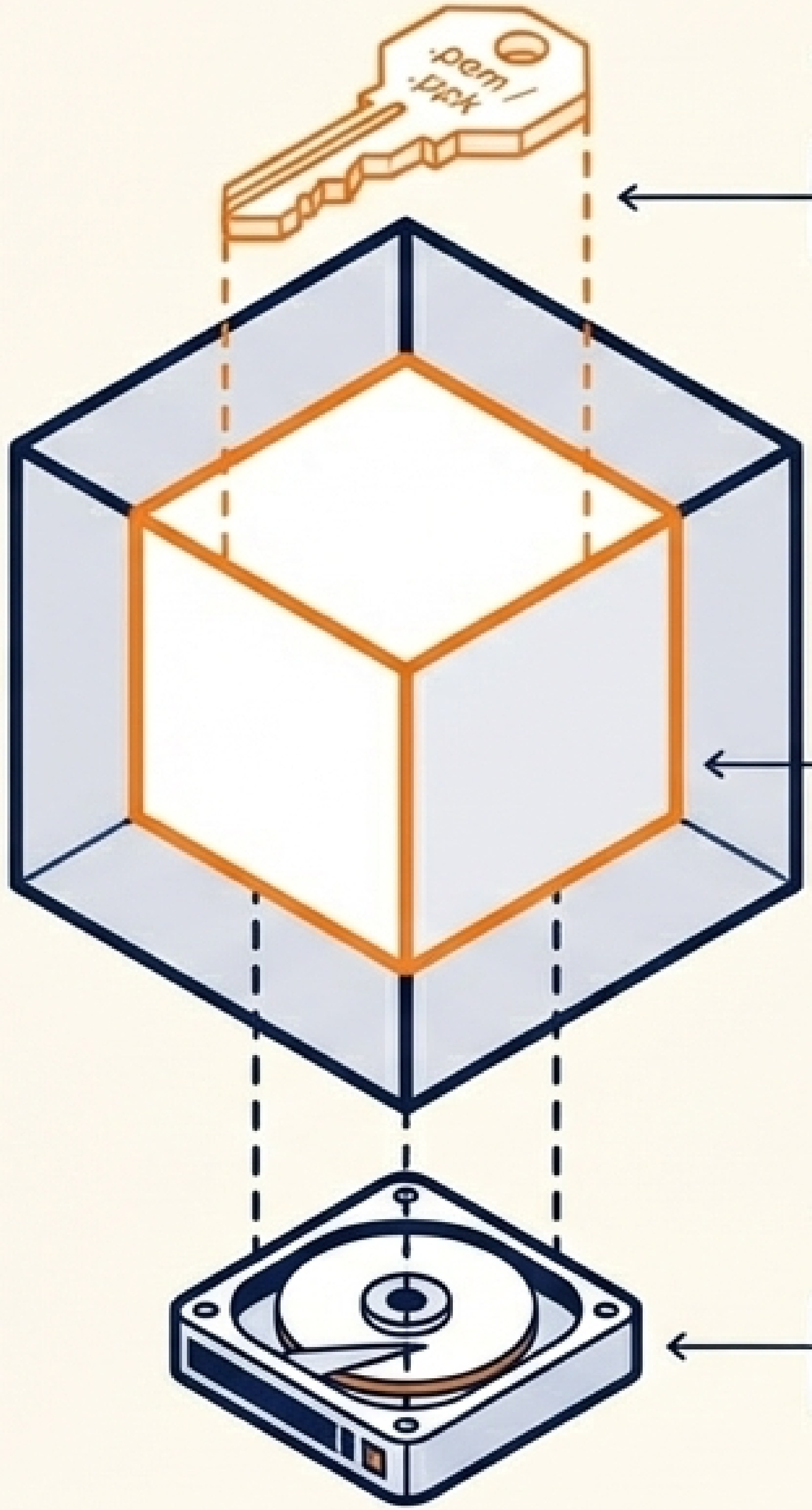
Anatomy of an EC2 Instance

The Key Pair (SSH Auth)

Required to securely remote-access the server's terminal from a local machine.

The Shield (Security Group)

The virtual firewall determining exactly who and what can communicate with the server.



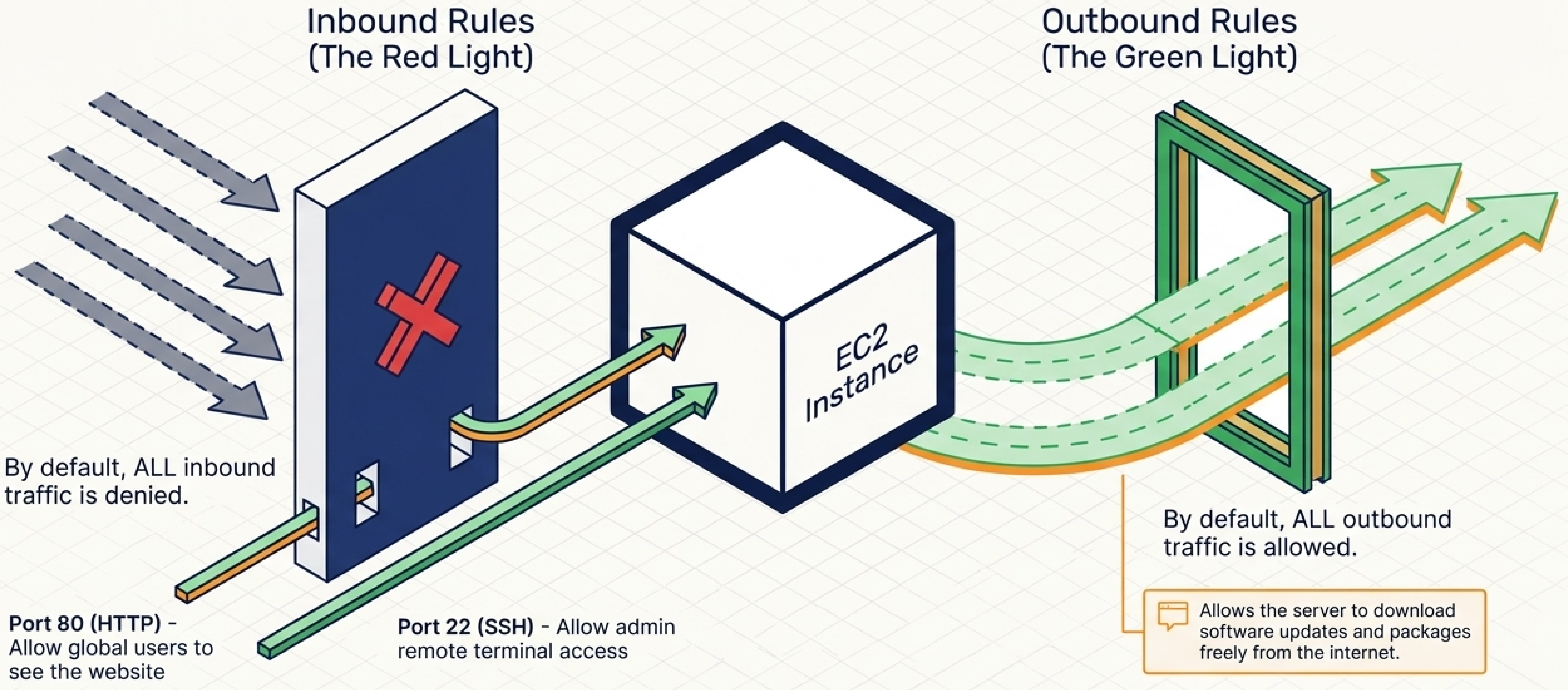
The Core (Compute)

Defines the Instance Type (CPU/RAM). e.g., t3.micro for testing, m5.large for web servers, g5.xlarge for GPU rendering.

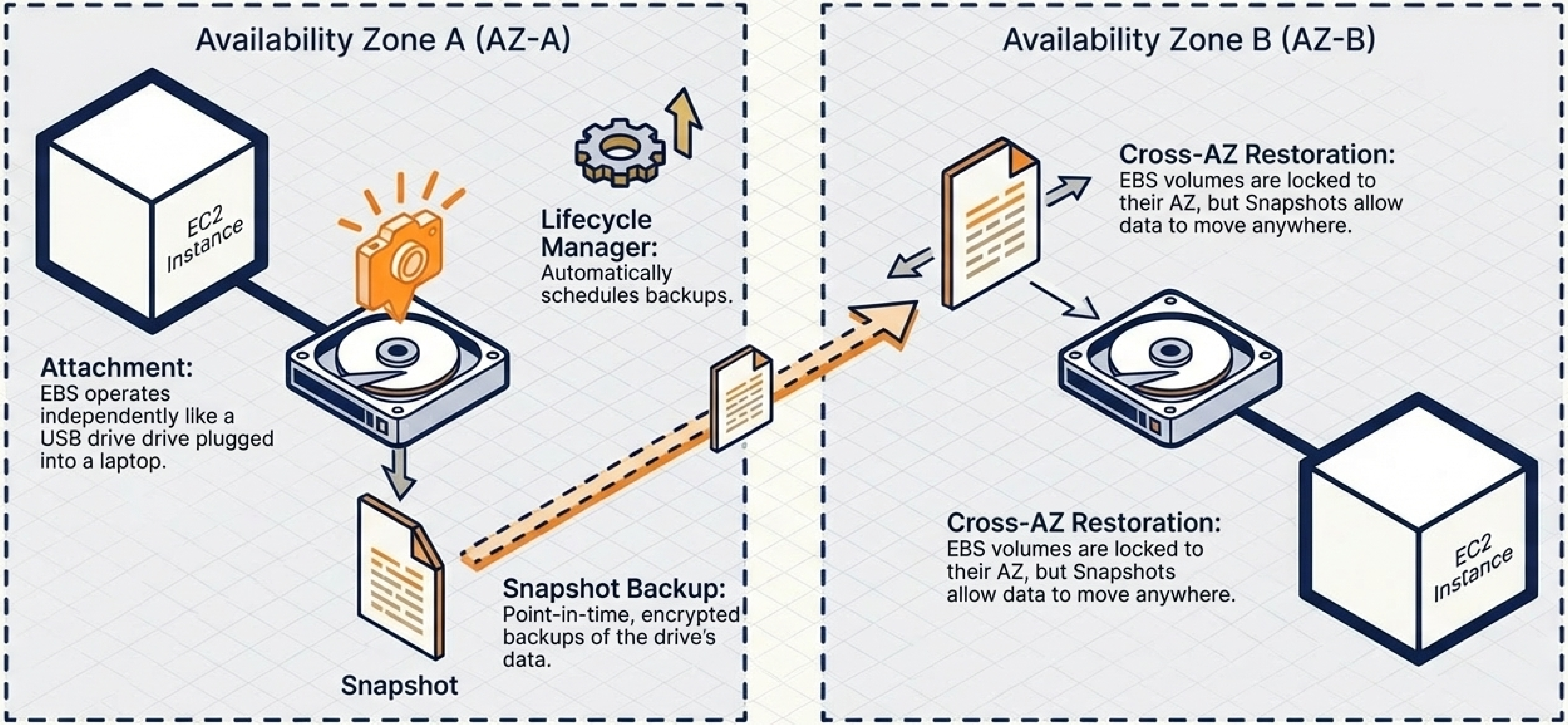
The Drive (EBS Volume)

Elastic Block Store. The virtual hard drive persisting data even if the instance stops.

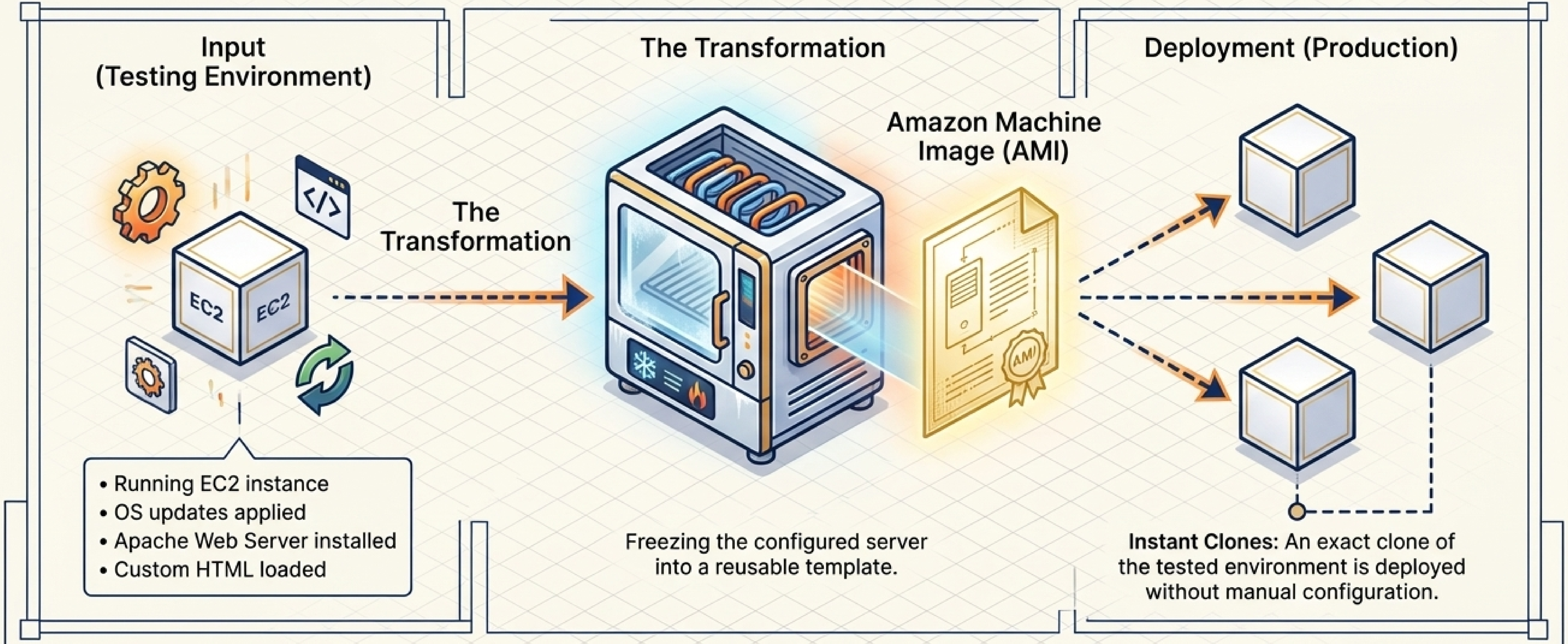
Security Groups: The Virtual Bouncer



The Storage Lifecycle: Elastic Block Store



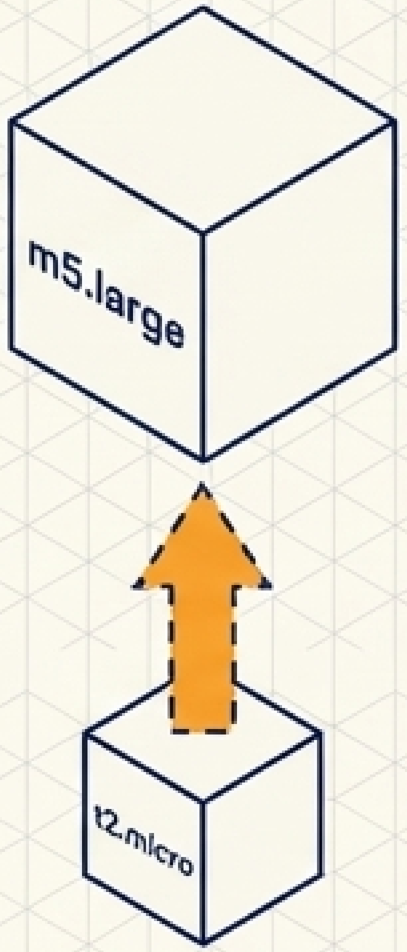
Freezing the Template: AMIs



Key Insight: AMIs eliminate repetitive manual configuration, ensuring consistency across fleets.

Scaling Strategies: Up vs. Out

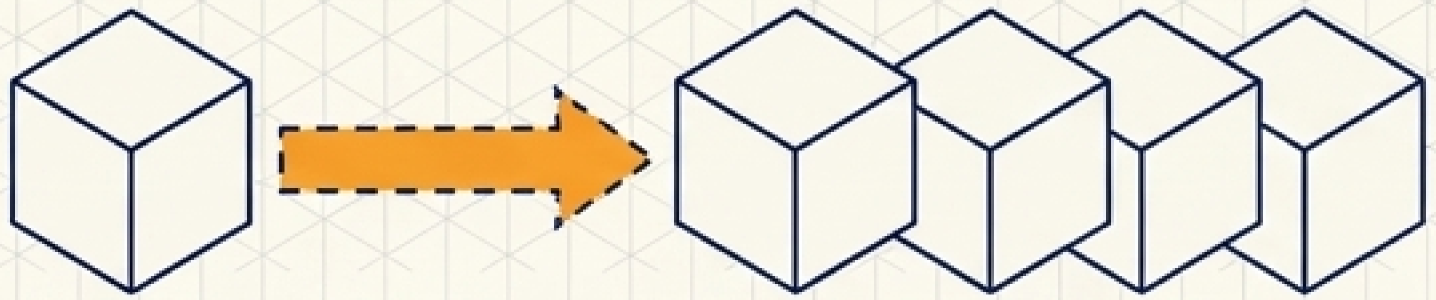
Vertical Scaling (Scaling Up)



Mechanism: Upgrading the instance type to give the same single server more CPU and RAM.

Limitation: Hard limits exist. You must eventually stop the server to upgrade it, causing system downtime.

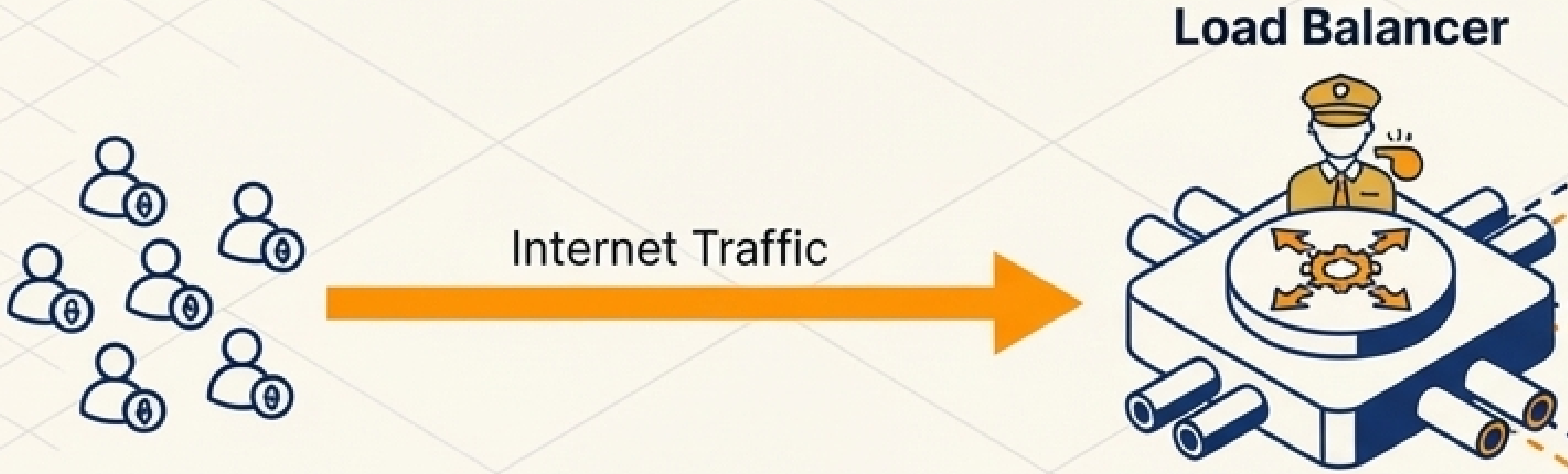
Horizontal Scaling (Scaling Out)

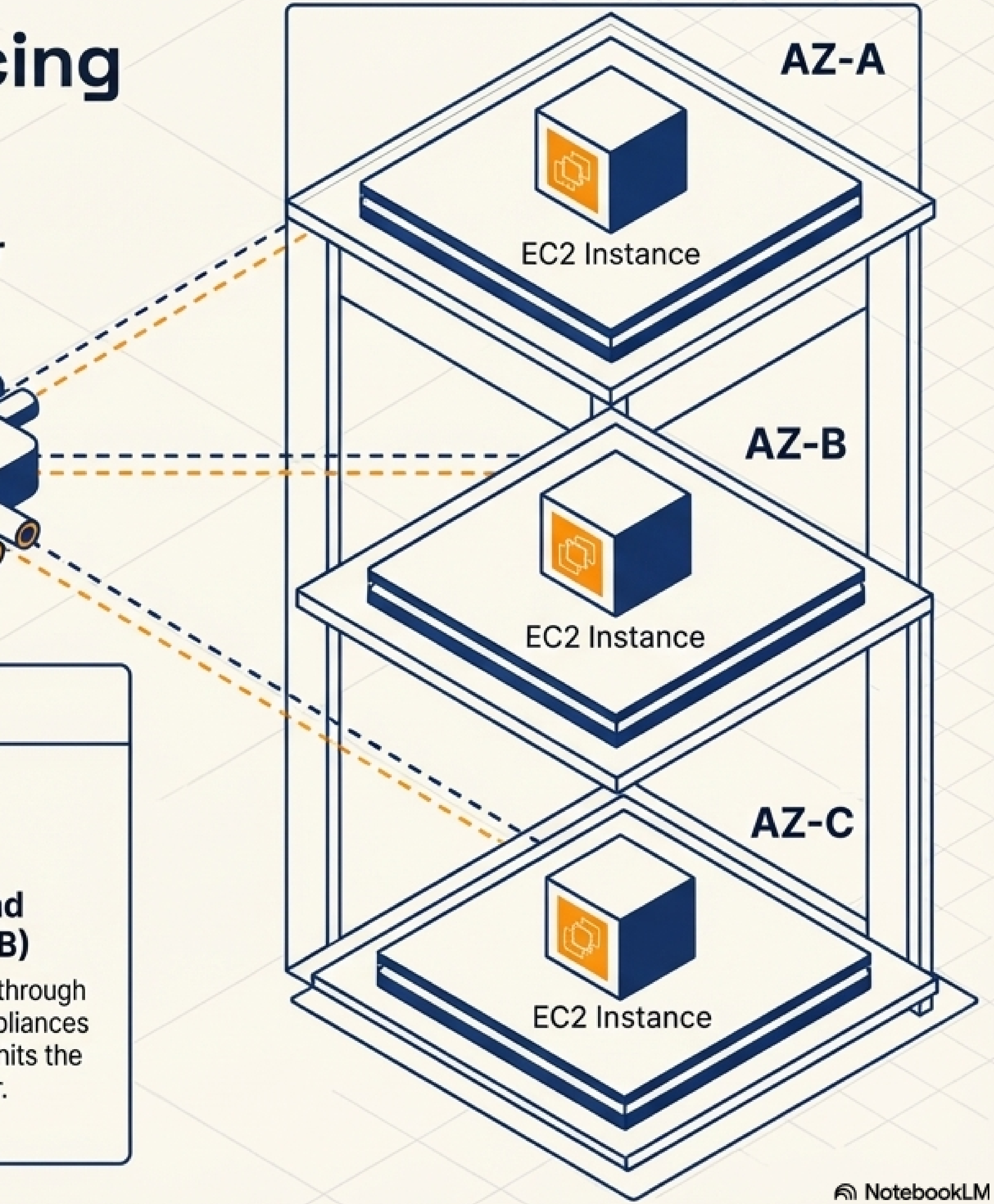


Mechanism: Adding more identical EC2 instances to share the incoming workload.

Advantage: Practically infinite scalability and built-in High Availability. If one instance fails, the others take the load.

The Traffic Director: Load Balancing





Types of ELB

Application Load Balancer (ALB)

Operates at Layer 7 (HTTP/HTTPS). Ideal for modern web applications and routing based on website content.

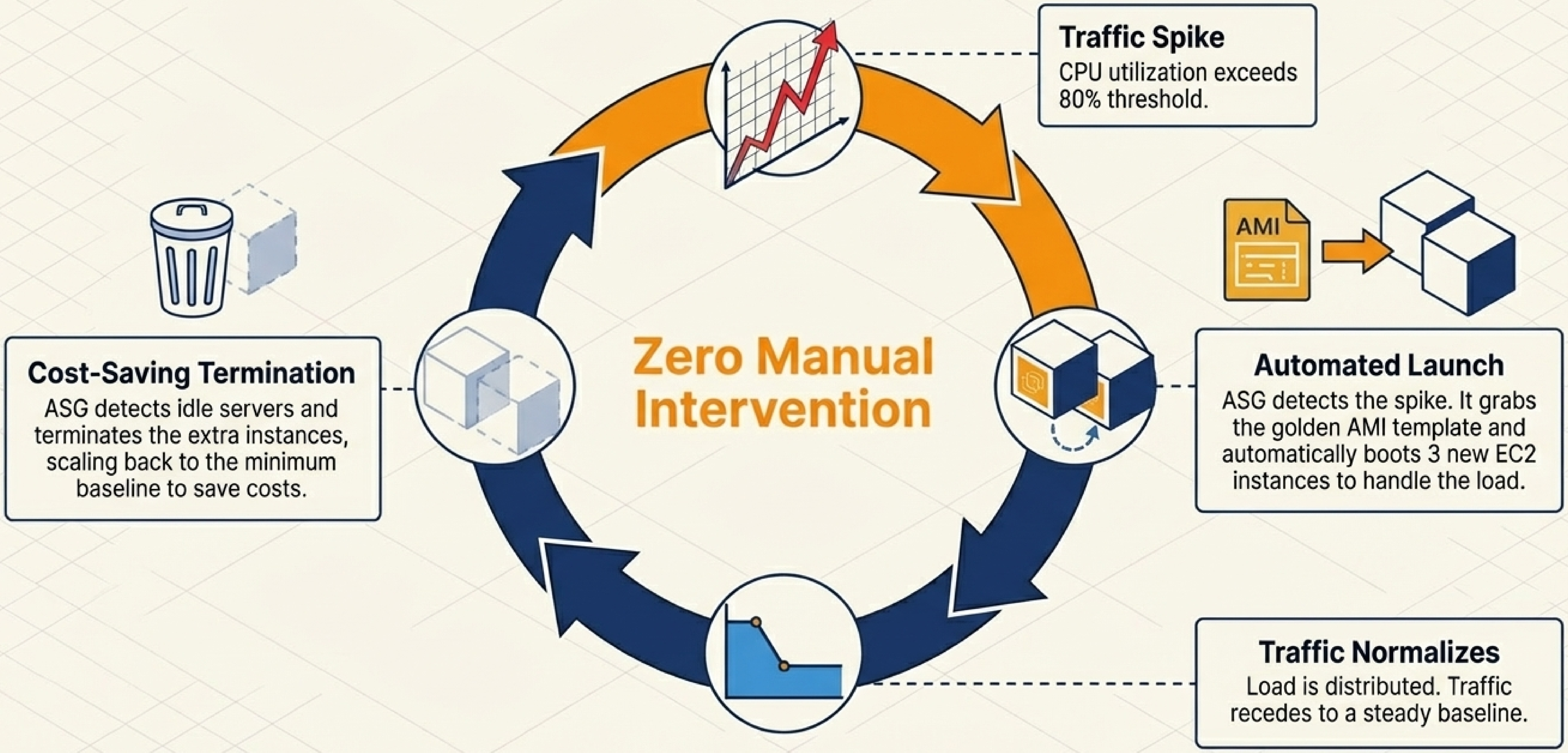
Network Load Balancer (NLB)

Operates at Layer 4 (TCP/UDP). Built for extreme performance, millions of requests per second, and ultra-low latency (Gaming, Finance).

Gateway Load Balancer (GLB)

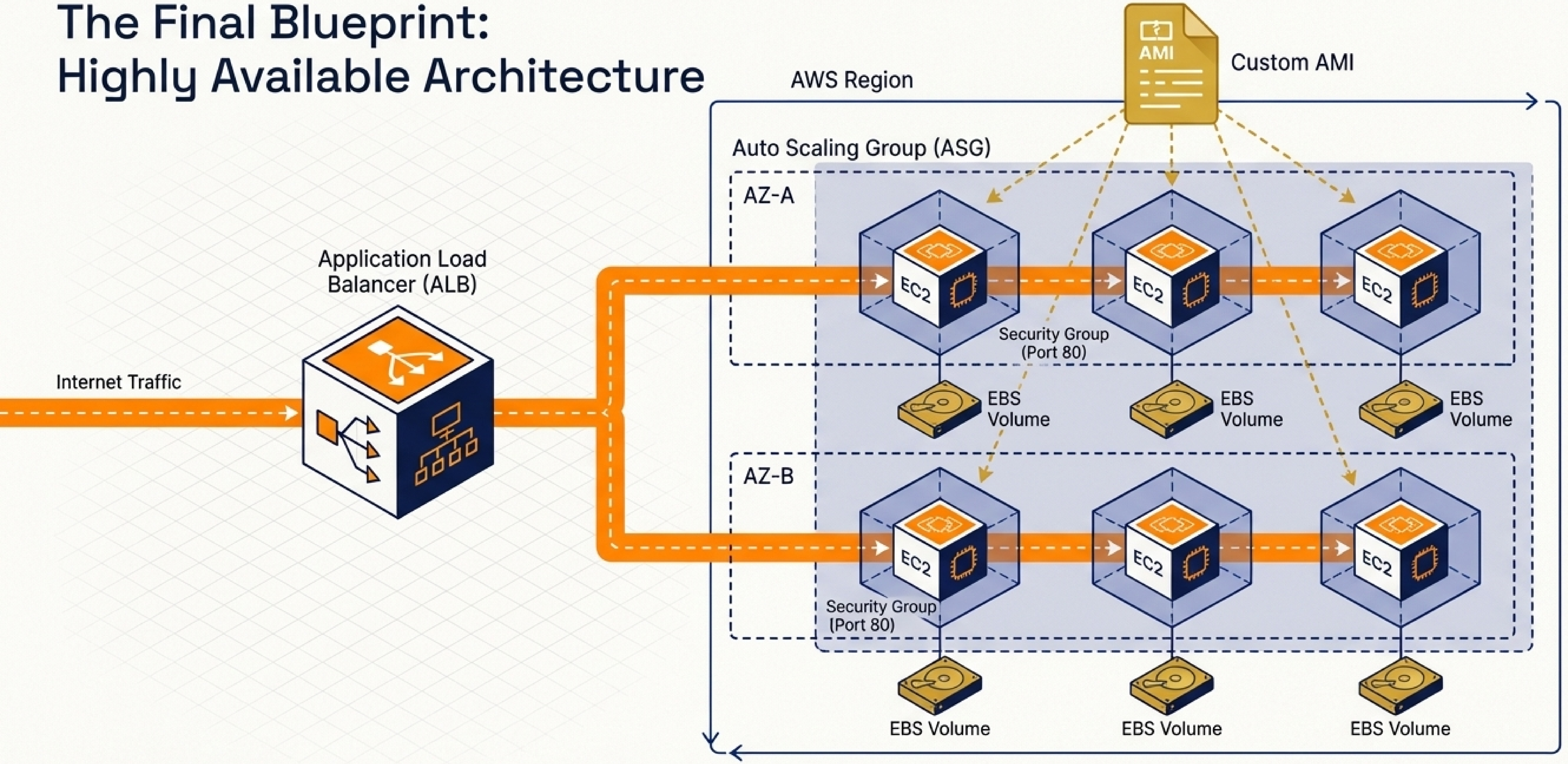
Built for routing traffic through third-party security appliances and firewalls before it hits the application layer.

Elasticity in Motion: Auto Scaling Groups



NotebookLM

The Final Blueprint: Highly Available Architecture



By combining single primitives, we achieve a self-healing, fault-tolerant, and elastic infrastructure capable of handling global demand.